

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

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Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

ANNEXURE A

INDUSTRY PROBLEM TASK

Diabetes Prediction and Reinforcement Learning-Based Treatment Recommendation

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general patterns from previously observed examples and uses those patterns to make predictions for unseen data.

For the given healthcare dataset, the suitable approach is supervised inductive learning, because the historical patient records contain input features and a known diabetes diagnosis/target variable.

Target Variable

The target variable is Diabetes / Outcome:

- 0 → Non-diabetic
- 1 → Diabetic

Important Features

1. Glucose Level – Glucose is one of the strongest indicators of diabetes. Higher glucose levels are generally associated with a higher probability of diabetes.
2. BMI – BMI indicates whether a person is underweight, normal weight, overweight, or obese. Higher BMI can be associated with increased risk of type 2 diabetes.
3. Age – Diabetes risk generally increases with age.

Other useful features include blood pressure, cholesterol, and family history.

1(b) Handling Class Imbalance

The dataset is assumed to contain more non-diabetic patients than diabetic patients. This creates a class imbalance problem.

Recommended Strategy: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) can be applied to the training data. SMOTE generates synthetic examples of the minority class rather than simply duplicating existing diabetic records.

Procedure:

1. Split the dataset into training and testing sets.
2. Apply preprocessing using the training data.
3. Apply SMOTE only to the training set.
4. Train the model using the balanced training data.
5. Evaluate on the original, untouched test set.

SMOTE should not be applied before the train-test split because this can cause data leakage.

1(c) 70:30 Train-Test Split

The dataset can be divided into 70% training data and 30% testing data. A stratified split should be used so that the diabetic/non-diabetic ratio is approximately maintained in both sets.

The 70:30 split provides sufficient data for training while retaining a relatively large independent test set. For a real clinical application, cross-validation and external validation using an independent clinical dataset would also be recommended.

Preprocessing:

- Numerical features such as age, blood pressure, cholesterol, glucose, and BMI can be standardized.
- Categorical features such as family history can be binary encoded or one-hot encoded.

Standardization:

$$z = (x - \mu) / \sigma$$

Normalization is particularly important for models such as Logistic Regression. Decision Trees generally do not require normalization.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree classifier predicts whether a patient is diabetic by repeatedly selecting features that best separate diabetic and non-diabetic patients.

Information Gain:

$$H(S) = -\sum p_i \log_2(p_i)$$

$$IG(S,A) = H(S) - \sum_v (|S_v|/|S|) H(S_v)$$

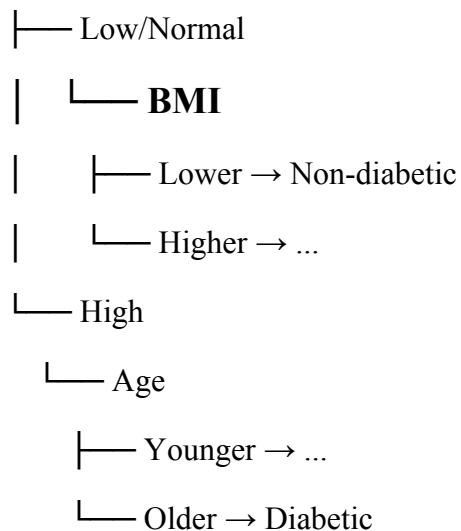
Gini Index:

$$Gini(S) = 1 - \sum p_i^2$$

The actual root feature should be obtained from the trained dataset. For example, if glucose provides the highest Information Gain, the tree could begin with glucose as the root.

Illustrative first three levels:

Glucose



The exact tree must be generated from the actual dataset.

Example Python:

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
```

```
dt = DecisionTreeClassifier(  
    criterion="gini",  
    random_state=42,  
    class_weight="balanced"  
)  
dt.fit(X_train, y_train)
```

```
plot_tree(  
    dt,  
    feature_names=X.columns,  
    class_names=["Non-Diabetic", "Diabetic"],  
    max_depth=2,  
    filled=True  
)
```

2(b) Model Evaluation

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

A False Negative occurs when an actually diabetic patient is predicted as non-diabetic. False negatives are especially important because they may result in delayed diagnosis and treatment.

To reduce false negatives:

- **SMOTE**
- Class weights
- Classification-threshold adjustment
- Optimizing recall or F2-score
- Increasing the penalty for false negatives
- Collecting more representative diabetic samples

For clinical deployment, high recall/sensitivity should receive significant attention.

2(c) Post-Pruning

A fully grown decision tree can overfit the training data. Post-pruning removes branches that provide insufficient predictive benefit.

Cost-complexity pruning:

$$R\alpha(T) = R(T) + \alpha|T|$$

where $R(T)$ is model error, $|T|$ is the number of terminal nodes, and α controls model complexity.

The report should compare:

Model | Accuracy | Precision | Recall | F1

Pre-pruned Tree | Actual value | Actual value | Actual value | Actual value

Post-pruned Tree | Actual value | Actual value | Actual value | Actual value

A post-pruned tree is generally preferable if it gives comparable or better performance with lower complexity. It is easier to interpret and audit in a clinical setting.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression predicts the probability that a patient has diabetes.

$$P(Y=1|X) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Example Python:

```
from sklearn.linear_model import LogisticRegression
```

```
lr = LogisticRegression(  
    class_weight="balanced",  
    max_iter=1000,
```

```
random_state=42  
)
```

```
lr.fit(X_train, y_train)
```

```
y_pred_lr = lr.predict(X_test)
```

```
y_prob_lr = lr.predict_proba(X_test)[:, 1]
```

The models should be compared using Accuracy and AUC-ROC.

Model	Accuracy	AUC-ROC
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Decision Tree	Actual value	Actual value
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Logistic Regression	Actual value	Actual value
---------------------	--------------	--------------

Logistic Regression is preferable when interpretability, probability estimates, and a simpler statistical model are important.

3(b) Feature Importance Analysis

Decision Tree importance can be obtained using:

```
dt.feature_importances_
```

For Logistic Regression, the absolute values of standardized coefficients can be used to rank features:

```
abs(lr.coef_[0])
```

The top three predictors from both models should be reported:

Rank | Decision Tree | Logistic Regression

1 | Actual feature | Actual feature

2 | Actual feature | Actual feature

3 | Actual feature | Actual feature

The models may agree on important predictors such as glucose or BMI, but they do not necessarily have identical rankings because Decision Trees capture nonlinear relationships while Logistic Regression models a linear relationship in the log-odds.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) MDP Formulation

The treatment recommendation problem can be formulated as a Markov Decision Process (MDP):

(S, A, P, R, γ)

where S = states, A = actions, P = transition probabilities, R = rewards, and γ = discount factor.

States

- S_0 = Low-risk / Healthy
- S_1 = Prediabetic

- S2 = Newly diagnosed / Moderate risk
- S3 = High-risk / Poorly controlled
- S4 = Improved / Controlled

States should ideally be derived from clinically meaningful measurements such as glucose, BMI, blood pressure, cholesterol, and previous patient state.

Actions

- A0 = Diet
- A1 = Exercise
- A2 = Medication
- A3 = Monitor

Medication should represent a clinician-reviewed treatment recommendation rather than autonomous prescribing.

Reward Function

An example reward structure:

+10 = significant health improvement

+5 = moderate improvement

+1 = small improvement

0 = no meaningful change

-5 = health deterioration

-10 = serious deterioration

A possible reward equation is:

$$R(s,a,s') = \text{HealthScore}(s') - \text{HealthScore}(s)$$

Transition Dynamics

$P(s'|s,a)$ represents the probability of moving from the current patient state to a future state after an action.

In a real healthcare application, transition probabilities should be estimated from validated longitudinal clinical data rather than invented.

4(b) Q-Learning

Q-Learning learns the expected long-term value of taking an action in a particular state.

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where α is the learning rate, γ is the discount factor, and r is the reward.

Example:

$$\alpha = 0.1$$

$$\gamma = 0.9$$

Example initial Q-table:

State | Diet | Exercise | Medication | Monitor

Low Risk | 0 | 0 | 0 | 0

Prediabetic | 0 | 0 | 0 | 0

Moderate Risk | 0 | 0 | 0 | 0

High Risk | 0 | 0 | 0 | 0

Controlled | 0 | 0 | 0 | 0

Example Q-table update 1:

If:

$Q(\text{Prediabetic}, \text{Diet}) = 0$

reward = 5

maximum future Q = 2

Then:

$$Q = 0 + 0.1[5 + 0.9(2) - 0]$$

$$Q = 0.68$$

Therefore:

$Q(\text{Prediabetic}, \text{Diet}) = 0.68$

Example Q-table update 2:

If:

Current Q = 0.68

Reward = 5

Maximum future $Q = 3$

Then:

$$Q = 0.68 + 0.1[5 + 0.9(3) - 0.68]$$

$$Q = 1.042$$

Therefore:

$$Q(\text{Prediabetic, Diet}) = 1.042$$

At least five patient episodes should be used for the experiment.

Example:

Episode 1 $\rightarrow S_0 \rightarrow S_1 \rightarrow S_4$

Episode 2 $\rightarrow S_1 \rightarrow S_2 \rightarrow S_4$

Episode 3 $\rightarrow S_2 \rightarrow S_3 \rightarrow S_4$

Episode 4 $\rightarrow S_1 \rightarrow S_4$

Episode 5 $\rightarrow S_3 \rightarrow S_4$

Convergence Criterion

A suitable convergence criterion is:

$$\max |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < \epsilon$$

For example, $\epsilon = 0.001$.

The algorithm can be considered converged when the maximum Q-value change remains below this threshold for a predefined number of consecutive iterations or episodes.

4(c) Final Learned Policy

The learned policy is:

$$\pi(s) = \operatorname{argmax}_a Q(s,a)$$

An illustrative policy could be:

Patient State | Best Action

Low Risk | Monitor

Prediabetic | Diet

Moderate Risk | Exercise

High Risk | Medication / Clinical Review

Controlled | Monitor

These are illustrative values. The final policy must be extracted from the learned Q-table and validated clinically.

Ethical Considerations

1. Patient Safety

The agent should not independently make high-risk medical decisions. Medication recommendations should require approval from qualified healthcare professionals.

2. Bias

If the training data underrepresents particular patient groups, the agent may produce poorer recommendations for those groups. The system should therefore be tested for fairness.

3. Explainability

Healthcare professionals should be able to understand why a recommendation was produced.

4. Human Oversight

The system should operate as a clinical decision-support system rather than an autonomous doctor. Clinicians should be able to review, reject, or override recommendations.

5. Data Privacy

Patient information should be protected through access controls, encryption, secure storage, audit logs, and data minimization.

6. Safe Exploration

Unrestricted RL exploration would be inappropriate with real patients because an unsafe action could cause harm. RL training should initially use simulated environments, retrospective datasets, or validated clinical simulators.

OVERALL CONCLUSION

The proposed system combines supervised learning and reinforcement learning.

The first stage uses patient data to predict diabetes risk. Data preprocessing, class balancing, and stratified train-test splitting improve model reliability.

A Decision Tree provides an interpretable diagnostic model, while Logistic Regression provides a statistical probability-based alternative. Their performance can be compared using Accuracy, Precision, Recall, F1-score, and AUC-ROC.

The second stage formulates personalized treatment recommendation as an MDP. Q-Learning can then learn the long-term value of different actions for different patient health states.

The RL component should not directly prescribe treatment. In a real healthcare environment, it should function as a clinician-supervised decision-support system with strong safety, fairness, privacy, and explainability mechanisms.

IMPORTANT NOTE

The numerical results requested in the assignment—Information Gain/Gini values, first three tree levels, Accuracy, Precision, Recall, F1, AUC-ROC, feature rankings, pruning comparison, Q-table values, and final policy—must be generated from the actual patient dataset. Inventing these values would make the analysis unreliable.

RECOMMENDED PYTHON LIBRARIES

```
import pandas as pd
```

```
import numpy as np
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
    classification_report
)
```

```
from imblearn.over_sampling import SMOTE
import matplotlib.pyplot as plt
```